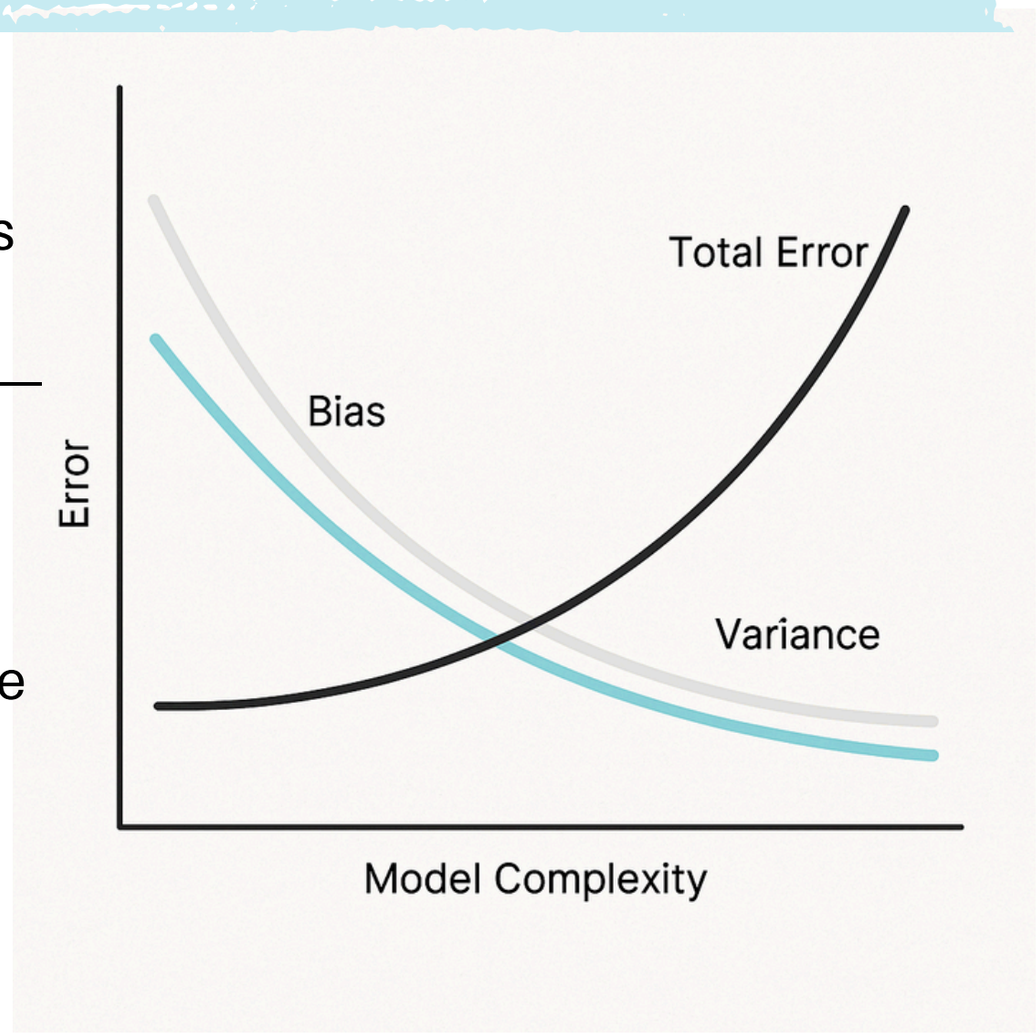
Bias–Variance Tradeoff

Understanding the bias-variance trade-off is one of the most important ideas in machine learning. It helps explain why a model makes errors and how we can control those errors when building better models.
When we train a model, we want it to perform well not only on the training data, but also on new, unseen data. However, there are two types of errors that can get in the way of this goal: bias and variance.

Bias refers to the error caused by simplifying the model too much. When a model has high bias, it makes assumptions about the data that are too strong. It can't capture the real patterns, so it ends up making basic, inaccurate predictions. This usually happens with models that are too simple — for example, using a straight line to fit a curved pattern. High bias leads to underfitting.
On the other hand, variance refers to how much the model's predictions change depending on the training data. A model with high variance fits the training data very closely even the noise or random errors — and it performs poorly on new data. This is what we call overfitting.



You can't minimize both bias and variance at the same time.
The goal is to find the balance that gives the lowest total error.

The challenge in machine learning is finding the right balance. If your model is too simple, you get high bias. If your model is too complex, you get high variance. The best model is somewhere in between — one that's complex enough to learn the patterns, but not so complex that it memorizes everything. This balance is what we call the bias-variance tradeoff. When we improve one, the other usually gets worse. So, we don't try to eliminate both we try to minimize total error, which includes both bias and variance.

In practice, we deal with this tradeoff by choosing the right algorithm, adjusting hyperparameters, and checking performance on validation data. For example, if we see our model is underfitting, we might reduce bias by making the model more flexible. If we see it's overfitting, we might reduce variance by using regularization or simplifying the model.
Understanding the bias-variance tradeoff helps us make smarter choices when training models. It reminds us that the goal isn't just to "fit the data" it's to generalize well to new data.